

Hotel Bookings Analytics Platform

End-to-End Data Analytics & Machine Learning Project Report

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Project type	Data Analytics & Machine Learning
Tech stack	Python 3.10+ Streamlit 1.63 Scikit-Learn 1.7.2 OpenAI GPT-4o-mini SQLite SHAP 0.49.1 pytest 8.4.2
Repository	github.com/chirag2005dv/hotel-bookings-analytics-platform
License	MIT
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1. Executive Summary

This report documents the Hotel Bookings Analytics Platform, a complete nine-phase, end-to-end data analytics and machine learning project built on the publicly available Hotel Booking Demand dataset (Antonio, Almeida & Nunes, 2019). The project transforms raw booking records into production-ready insights: data cleaning, exploratory analysis, SQL analytics, machine learning, model explainability, an interactive Streamlit dashboard, and AI-powered natural language insights via GPT-4o-mini.

Key outcomes achieved

Area	Outcome
Dataset	118,564 clean bookings (from 119,390 raw), July 2015 - August 2017
Core KPI	37.26% overall cancellation rate; City Hotel 41.9% vs Resort Hotel 28.0%
Best ML model	HistGradientBoosting: ROC-AUC 0.9157, F1 0.7832, Accuracy 83.6%
Top features	Deposit Type (#1 SHAP/Permutation/MDI), Lead Time (#2), Special Requests (#3)
Revenue	\$25.99M total estimated revenue; Online TA drives 52.8% of revenue
Dashboard	7-tab Streamlit dashboard with real-time ML prediction and GPT-4o-mini insights
Testing	73 automated pytest tests across 3 test files

2. Project Overview

2.1 Objectives

The primary business objective is to predict hotel booking cancellations (`is_cancelled`) at booking time, before check-in, using only features available at booking creation, enabling hotel operators to take proactive action. Secondary objectives include:

- Understand revenue trends and ADR patterns across hotel types, years, and segments
- Identify high-risk customer segments for targeted retention strategies
- Surface actionable operational insights through 20 annotated SQL queries
- Provide hotel managers with an interactive self-service analytics dashboard
- Explain model predictions in plain language using SHAP and AI narration

2.2 Pipeline Architecture

Phase	Name	Description
Phase 1	Project Setup	Repository initialisation, dependency pinning, .env configuration
Phase 2	Data Cleaning	NULL handling, dtype casting, invalid row removal, outlier flagging, feature engineering
Phase 3	EDA & KPIs	33 charts across 8 analytical sections, structured KPI report
Phase 4	SQL Analytics	20 annotated SQL queries via SQLite, cross-validated against Phase 3
Phase 5	Machine Learning	Binary classification pipeline: 3 models, cross-validation, evaluation
Phase 6	Explainability	SHAP TreeExplainer, Permutation Importance, Partial Dependence Plots

Phase	Name	Description
Phase 7	Dashboard	7-tab Streamlit dashboard with global sidebar filters and Plotly charts
Phase 8	AI Insights	GPT-4o-mini integration with data-grounded natural language analysis
Phase 9	Testing	73 automated pytest tests; synthetic in-memory data, no network calls

3. Dataset Description

The Hotel Booking Demand dataset contains one row per hotel booking, covering a City Hotel and a Resort Hotel in Portugal.

Attribute	Value
Source file	hotel_bookings.csv
Publication	Antonio, Almeida & Nunes (2019) - ScienceDirect / Kaggle
Raw rows / cols	119,390 rows x 32 columns
Cleaned rows / cols	118,564 rows x 42 columns (including 7 engineered features)
Target variable	is_canceled (0 = Not Canceled, 1 = Canceled)
Class balance	74,388 Not Canceled (62.7%) 44,176 Canceled (37.3%)

3.1 Engineered Features (Phase 2 Additions)

Feature	Definition
total_nights	stays_in_weekend_nights + stays_in_week_nights
arrival_date	Parsed from year + month + day_of_month
room_type_match	1 if reserved_room_type == assigned_room_type
is_high_season	1 if arrival month is June, July, or August
revenue_estimate	adr * total_nights
total_guests	adults + children + babies
arrival_month_num	Numeric month 1-12 for time-series ordering

4. File Structure

The project is organised into a professional folder hierarchy:

```
datasets/
|-- src/           # Pipeline scripts (data_cleaning, eda, sql, ml, dashboard)
|-- audits/        # Phase-specific audit and validation scripts
|-- data/          # Raw and processed datasets (gitignored)
|-- sql/           # Annotated business queries (.sql)
|-- reports/       # Output logs, KPI summaries, markdown decisions
|-- models/        # Serialised ML models (.pkl)
|-- plots/         # 33 EDA and ML evaluation chart images (PNG)
|-- screenshots/   # Dashboard UI screenshots
|-- sql_results/   # CSV outputs from all 20 SQL queries
|-- tests/         # pytest suite (73 tests)
|-- pytest.ini     # Test configuration
|-- requirements.txt # Pinned dependencies
`-- README.md      # Main project documentation
```

5. Phase 2 - Data Cleaning & Preprocessing

Script: src/data_cleaning.py | Docs: reports/PREPROCESSING_DECISIONS.md

The cleaning pipeline is a 10-step, sequential, idempotent process. Key steps:

- 1. **NULL strings:** 129,421 occurrences of the literal 'NULL' string converted to true NaN.
- 2. **Missing values:** country filled with 'Unknown'; agent/company with 0; children with 0.
- 3. **Invalid rows:** removed zero-night (715), zero-guest (110), and negative-ADR (1) bookings.
- 4. **Leakage columns:** reservation_status, reservation_status_date, revenue_estimate, and engineered flags were carefully excluded from ML training sets.

6. Phase 3 - Exploratory Data Analysis & KPIs

Script: src/eda_analysis.py | Output: reports/kpi_report.txt | Charts: plots/

6.1 Core KPIs

KPI	Value
Overall cancellation rate	37.26%
Average Daily Rate (ADR)	\$101.01 (City Hotel: \$106.62 Resort Hotel: \$92.01)
Total revenue estimate	\$25,986,976 (checked-out bookings only)
Average lead time	104.5 days (Canceled: 145.0 d Not Canceled: 80.5 d)

6.2 Cancellation Profiles

- **By deposit type:** Non Refund (99.4%), No Deposit (28.6%), Refundable (22.2%)
- **By lead time:** monotonic increase from 11.0% (0-7 days) to 67.7% (365+ days)
- **By segment:** Groups (61.2%), Online TA (36.9%), Direct (15.5%)

7. Phase 4 - SQL Analytics

Query library: sql/hotel_queries.sql | Runner: src/sql_analytics.py

Twenty annotated business queries were executed via in-memory SQLite, yielding the following key insights:

Query	Insight
Q04 (Deposits)	Non-Refund deposits have a 99.4% cancel rate, the strongest single-feature predictor in the dataset.
Q07 (Segments)	Online TA drives 47.3% of volume and 52.8% of total revenue.
Q10 (Trends)	ADR grew 24.7% from 2015 (\$89.84) to 2017 (\$112.02).
Q14 (Loyalty)	Repeat guests cancel 2.4x less (15.7% vs 37.9%) and make more special requests.

8. Phase 5 - Machine Learning Pipeline

Script: src/ml_pipeline.py | Report: reports/ml_report.txt

- **Task:** binary classification to predict is_canceled at booking time.
- **Features:** 31 features (22 numeric scaled, 9 categorical encoded) after removing 7 leakage variables.
- **Best model:** HistGradientBoostingClassifier (ROC-AUC **0.9157**, Accuracy **83.6%**, F1 0.7832).

Top 5 Features (Mean Decrease in Impurity)

Rank	Feature	Importance
1	Deposit Type	20.9%
2	Lead Time	12.9%
3	Special Requests	10.0%
4	Market Segment	7.9%
5	Room Type Match	7.2%

9. Phase 6 - Model Explainability

Script: src/explainability.py

Three methods (SHAP TreeExplainer, Permutation Importance, and MDI) were applied to establish a consensus ranking of feature importance:

1. **Deposit Type:** ranks #1 across all methods. Non-Refundable rates reflect pricing policy, not direct causation.
2. **Lead Time:** ranks #2 across all methods. Longer lead time means higher probability of cancellation.
3. **Total Special Requests:** ranks #3 across all methods. Acts as a strong negative indicator (more requests, less likely to cancel).

10. Phase 7 & 8 - Dashboard & AI Insights

Script: `src/dashboard.py` and `src/ai_insights.py`

A comprehensive 7-tab Streamlit dashboard provides the following main capabilities:

1. **Overview & KPI panels:** interactive metrics and charts filtered globally by Hotel, Year, and Segment.
2. **Cancellation & Revenue tabs:** deep-dives into booking profiles and ADR trends.
3. **ML Prediction:** live form generating a real-time cancellation probability using the trained HistGBM model.
4. **Explainability:** interactive SHAP visualisations (Beeswarm, Summary, Dependence).
5. **AI Insights (GPT-4o-mini):** natural language narration dynamically grounded only in validated SQL and ML reports (zero-hallucination design).

11. Phase 9 - Testing

Framework: `pytest 8.4.2` | Tests: `tests/`

- **73 automated tests** covering data cleaning, SQL logic, ML evaluation integrity, and AI context assembly.
- Uses entirely in-memory synthetic data to prevent side-effects on production artifacts.

12. Limitations & Caveats

1. **Deposit artefact:** the 99.4% cancel rate on non-refundable rates likely reflects platform (OTA) behaviours and policies, requiring business validation before relying heavily on this feature for intervention.
2. **Partial date range:** the dataset lacks full calendar coverage for 2015 and 2017, introducing slight seasonal bias into year-over-year comparisons.
3. **Identity linkage:** true guest-level tracking is unavailable; repeat guest statistics rely purely on the PMS flag.

13. Setup & Usage Guide

```
# Clone and setup
git clone https://github.com/chirag2005dv/hotel-bookings-analytics-platform.git
pip install -r requirements.txt

# (Optional) Add OpenAI key to .env for AI insights
copy .env.example .env

# Run pipeline phases
python src/data_cleaning.py
python src/eda_analysis.py
python src/sql_analytics.py
python src/ml_pipeline.py
python src/explainability.py

# Launch dashboard
streamlit run src/dashboard.py

# Run tests
python -m pytest tests/ -v
```

14. Conclusion

The Hotel Bookings Analytics Platform demonstrates a production-quality data science workflow on a real-world hospitality dataset. Through 118,564 clean records, 33 charts, 20 queries, and a robust ML model (ROC-AUC 0.9157), the project highlights *Deposit Type*, *Lead Time*, and *Special Requests* as the strongest predictors of cancellation.

The modular architecture (clean, explore, query, model, explain, dashboard) provides hotel operators with an intuitive, interactive tool to assess booking risks dynamically, backed by a fully tested and extensible codebase.

References

- Antonio, N., de Almeida, A., & Nunes, L. (2019). Hotel booking demand datasets. *Data in Brief*.
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *NeurIPS*.
- Scikit-Learn, Streamlit, and OpenAI documentation.